

ECO102: Introduction to Macroeconomics

Lecture 10: The Loanable Funds Market Contd. *(Chapter 24: Finance, Saving, and Investment)*

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Chapter 24: Finance, Saving, and Investment, pp 575 – 580

Changes in The Demand for Loanable Funds

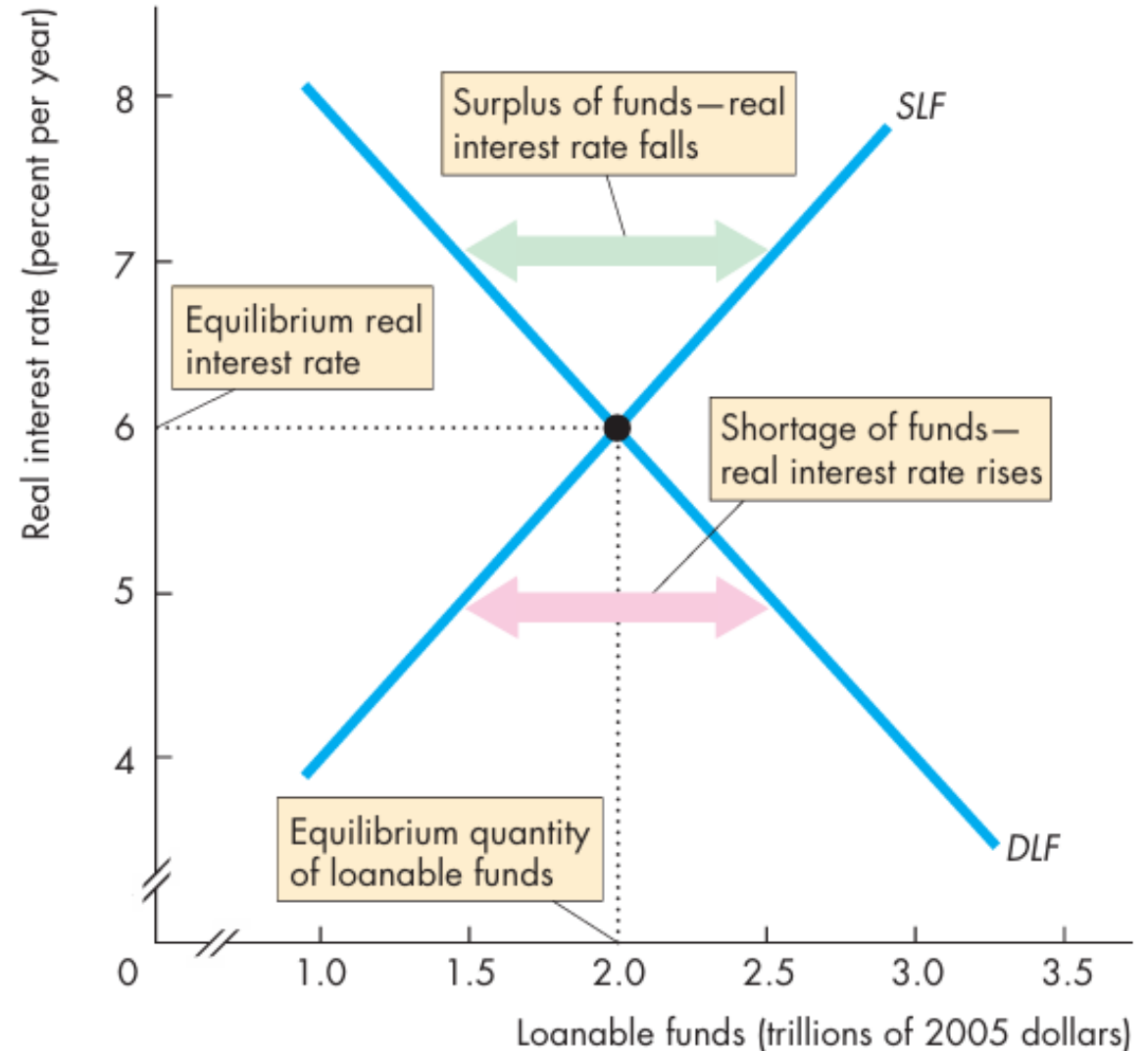
- When the expected profit changes, the demand for loanable funds changes
- Other things remaining the same, the **greater** the expected profit from new capital, the **greater** is the amount of investment and the greater the demand for loanable funds, **causing the demand for loanable curve to shift to the right**
- Expected profit can:
 - rise during a business cycle expansion and fall during a recession
 - rise when technological change creates profitable new products
 - rise as a growing population brings increased demand for goods and services
 - fluctuate with contagious swings of optimism and pessimism

Changes in The Supply for Loanable Funds

- ***Expected Future Income:*** The higher a household's expected future income, other things remaining the same, the smaller is its saving today
- ***Disposable Income:*** When disposable income increases, other things remaining the same, consumption expenditure increases but by less than the increase in income
Some of the increase in income is saved leading to a greater quantity of loanable funds supplied, other things remaining the same
- ***Wealth:*** The higher a household's wealth, other things remaining the same, the smaller is its saving; . If a person's wealth increases because of a capital gain, the person sees less need to save
- ***Default Risk:*** The greater is the default risk, the higher is the interest rate needed to induce a person to lend and the smaller is the supply of loanable funds
- ***An increase in disposable income, a decrease in expected future income, a decrease in wealth, or a fall in default risk increases saving and increases the supply of loanable funds***

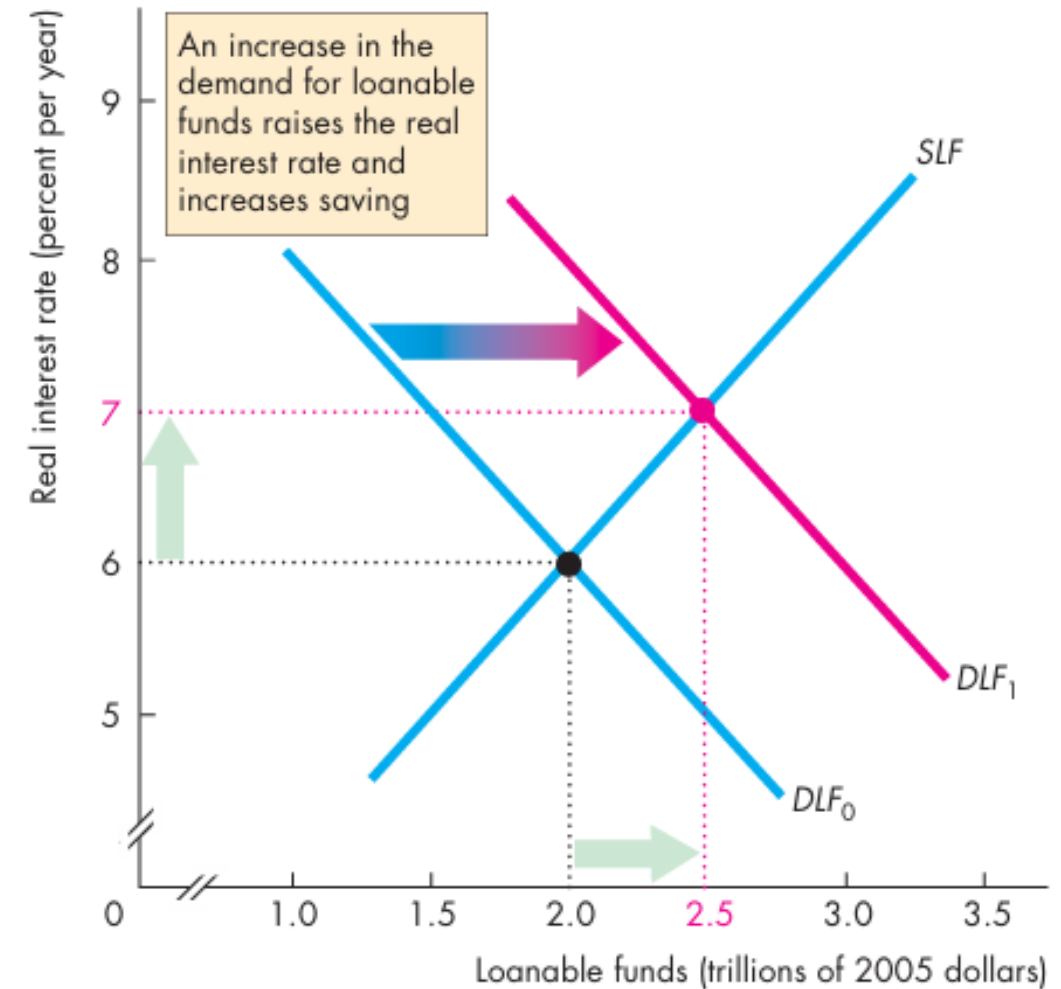
Equilibrium in the Loanable Funds Market

- If the real interest rate exceeds the equilibrium level; there is a surplus of funds
- The lenders are willing to take lower interest rates to reduce the surplus. The real interest rate falls and continues to fall until the quantity of funds supplied equals the quantity of funds demanded
- If the real interest rate is less than the equilibrium level; there is a shortage of funds
- The borrowers are willing to pay higher interest rate due to the shortage. So the real interest rate rises and continues to rise until the quantity of funds supplied equals the quantity demanded.



Changes in Demand

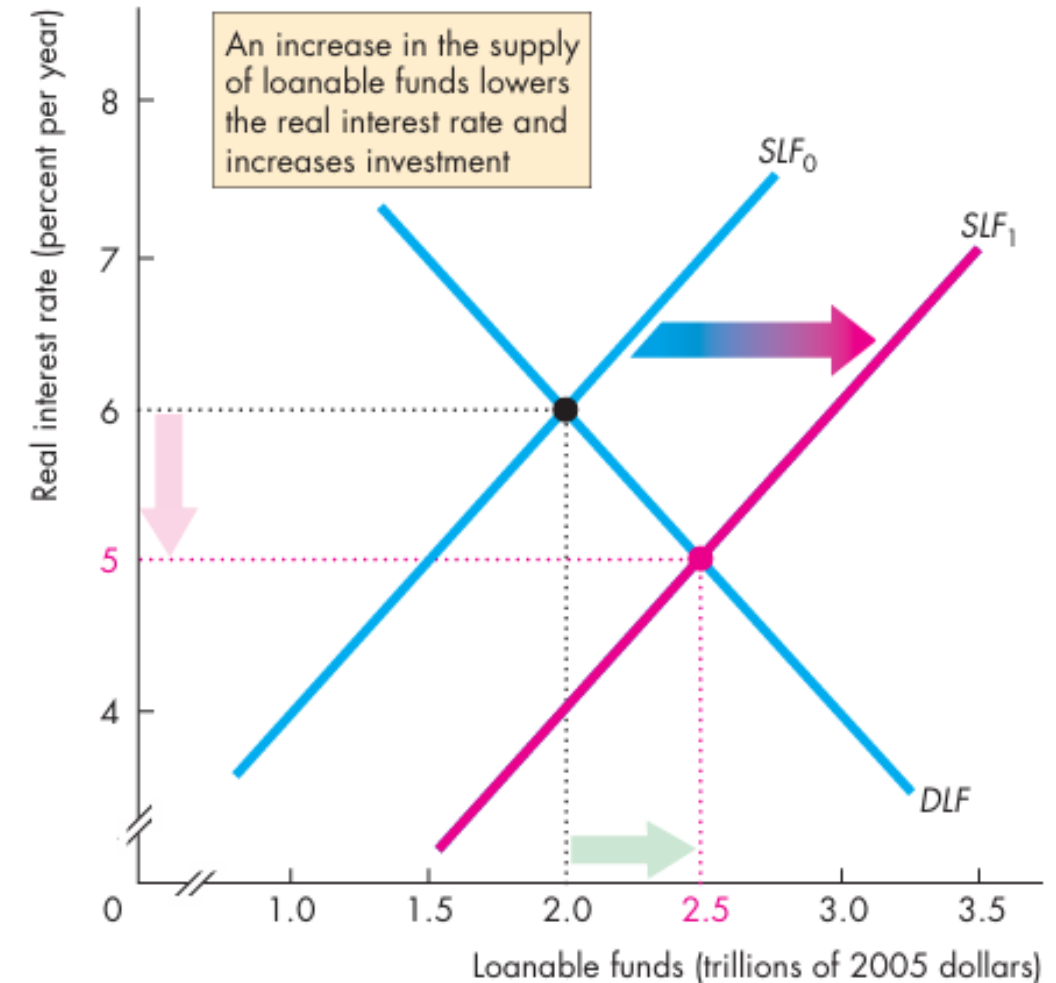
- If the expected future profit of firms increase, they increase their planned investment and increase their demand for loanable funds to finance that investment
- With an increase in the demand for loanable funds, but no change in the supply of loanable funds, there is a shortage of funds
- As borrowers compete for funds, the interest rate rises and lenders increase the quantity of funds supplied



(a) An increase in demand

Changes in Supply

- If one of the influences on saving plans changes and increases saving, the supply of loanable funds increases
- With no change in the demand for loanable funds, the market is flush with loanable funds
- Borrowers find bargains and lenders find themselves accepting a lower interest rate. At the lower interest rate, borrowers find additional investment projects profitable and increase the quantity of loanable funds that they borrow



(b) An increase in supply

Government in the Loanable Funds Market

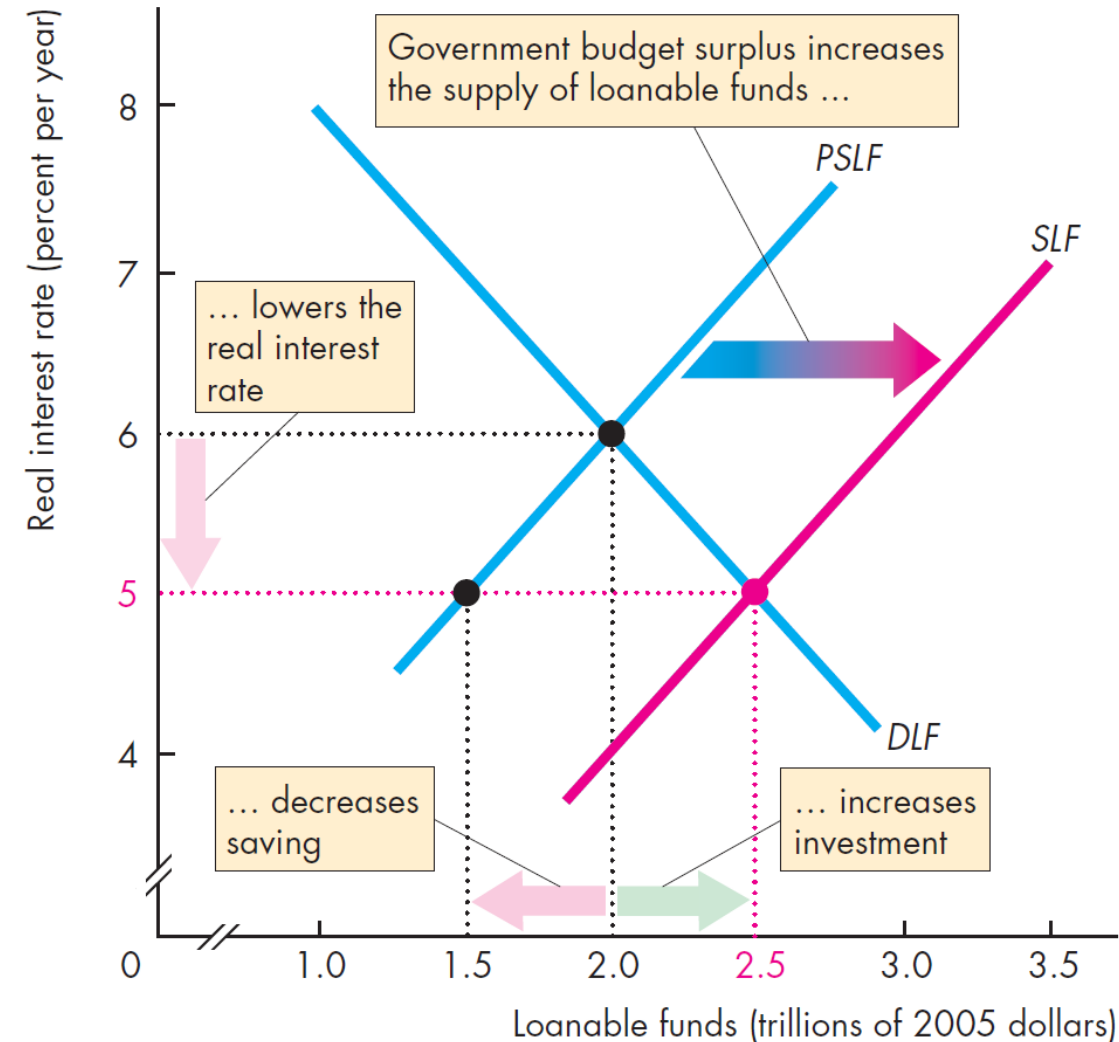
- A Government budget surplus *increases the supply* of loanable funds and contributes to financing investment
- A Government budget deficit *increases the demand* for loanable funds and competes with businesses for funds
- **Government Budget Surplus:** A Government Budget surplus shifts the supply of loanable funds curve to the right

The real interest rate falls, which decreases household saving and decreases the quantity of private funds supplied. The lower real interest rate increases the quantity of loanable funds demanded, and increases investment

Government in the Loanable Funds Market

A government budget surplus of \$1 trillion is added to private saving and the private supply of loanable funds (PSLF) to determine the supply of loanable funds, SLF

The real interest rate falls to 5 percent a year, private saving decreases, but investment increases to \$2.5 trillion



Government in the Loanable Funds Market

- **Government Budget Deficit:** A government budget deficit increases the demand for loanable funds, shifting the demand for loanable funds curve to the right.

The real interest rate rises, which increases household saving and increases the quantity of private funds supplied. But the higher real interest rate decreases investment and the quantity of loanable funds demanded by firms to finance investment

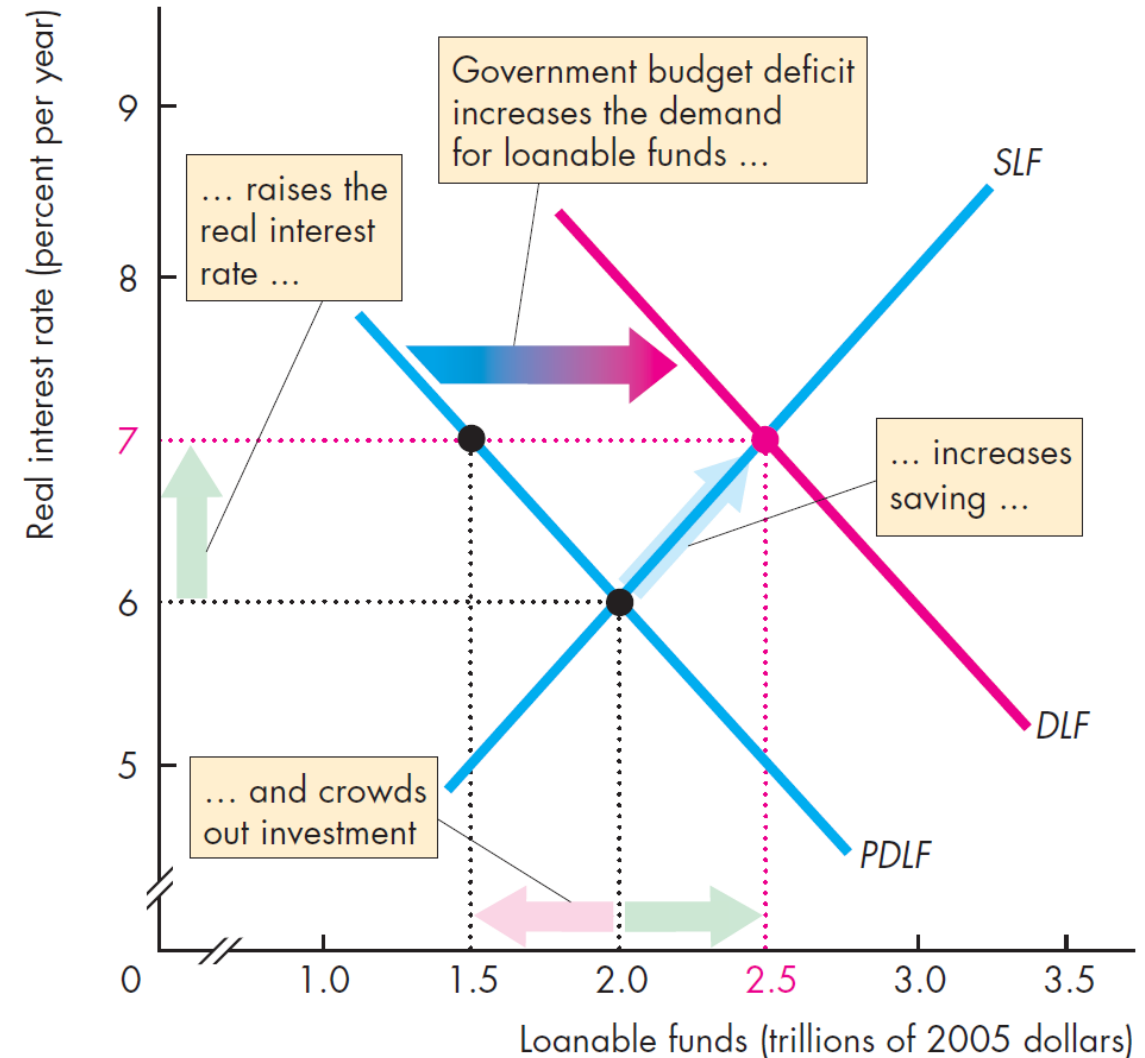
The tendency for a government budget deficit to raise the real interest rate and decrease investment is called ***the crowding-out effect***.

The crowding-out effect does not decrease investment by the full amount of the government budget deficit because the higher real interest rate induces an increase in private saving that partly contributes toward financing the deficit

Government in the Loanable Funds Market

A government budget deficit adds to the private demand for loanable funds curve (PDLF) to determine the demand for loanable funds curve, DLF

The real interest rate rises, saving increases, but investment decreases; a crowding-out effect



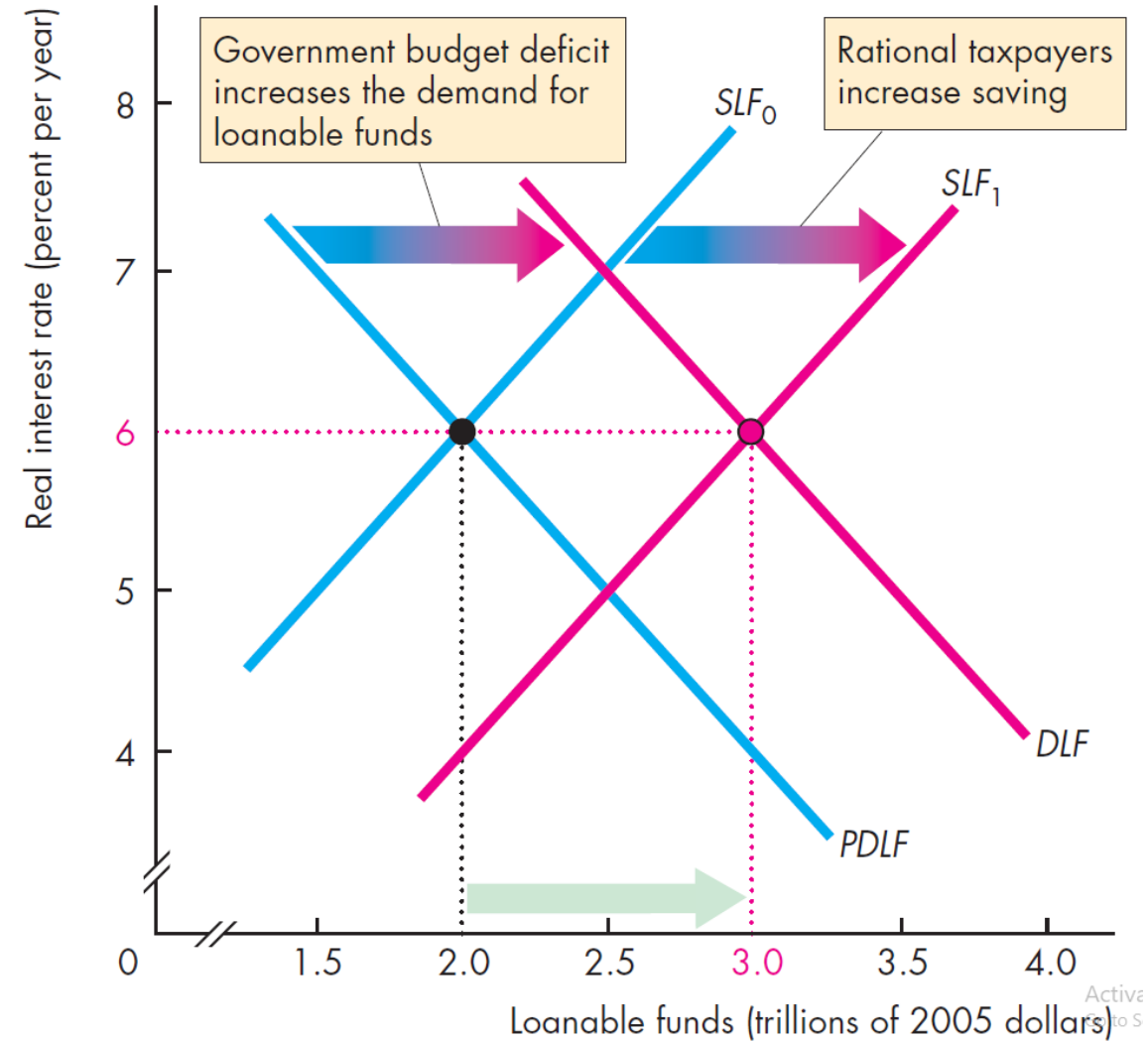
The Ricardo-Barro Effect

- First suggested by the English economist David Ricardo in the eighteenth century and refined by Robert J. Barro of Harvard University
- It states that the government budget, whether in surplus or deficit, has no effect on either the real interest rate or investment
- Barro says that taxpayers are rational; they assume a budget deficit today means that future taxes will be higher and future disposable incomes will be smaller, increasing savings today.
- Most economists regard the Ricardo-Barro view as extreme. But there might be some change in private saving that goes in the direction suggested by the Ricardo-Barro effect that lessens the crowding-out effect

The Ricardo-Barro Effect

A budget deficit increases the demand for loanable funds

Rational taxpayers increase saving, which increases the supply of loanable funds curve from SLF_0 to SLF_1 . Crowding out is avoided: Increased saving finances the budget deficit



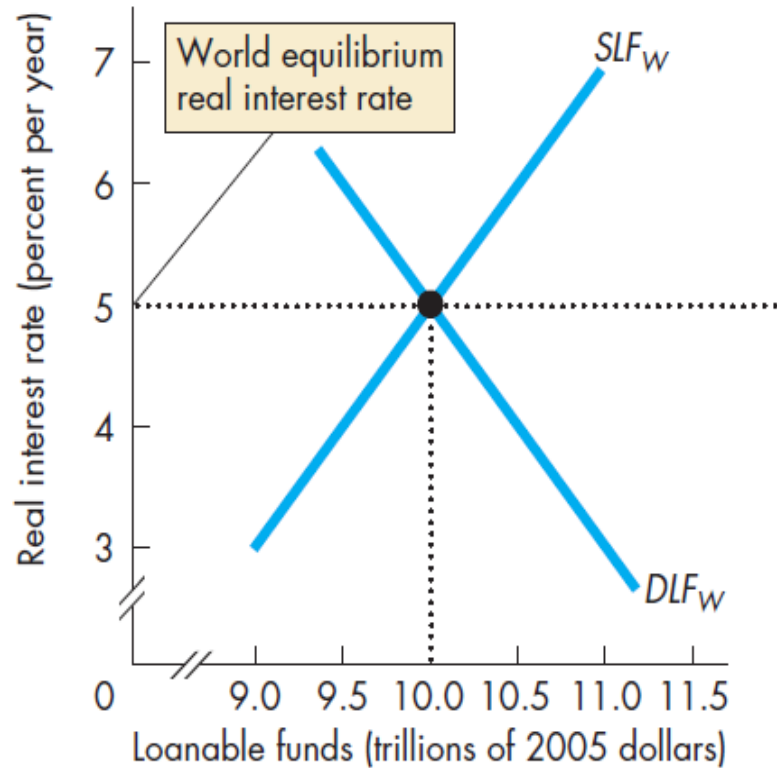
The Global Loanable Funds Market

- The loanable funds market is a single, integrated, global market. Funds flow into the country in which the interest rate is highest and out of the country in which the interest rate is lowest
- Lenders on the supply side of the market want to earn the highest possible real interest rate and they will seek it by looking everywhere in the world
- Borrowers on the demand side of the market want to pay the lowest possible real interest rate and they will seek it by looking everywhere in the world
- The interest rate on a risky loan minus that on a safe loan is called the ***risk premium***
- International capital mobility brings real interest rates in all parts of the world to equality except for differences that reflect differences in risk—differences in the risk premium

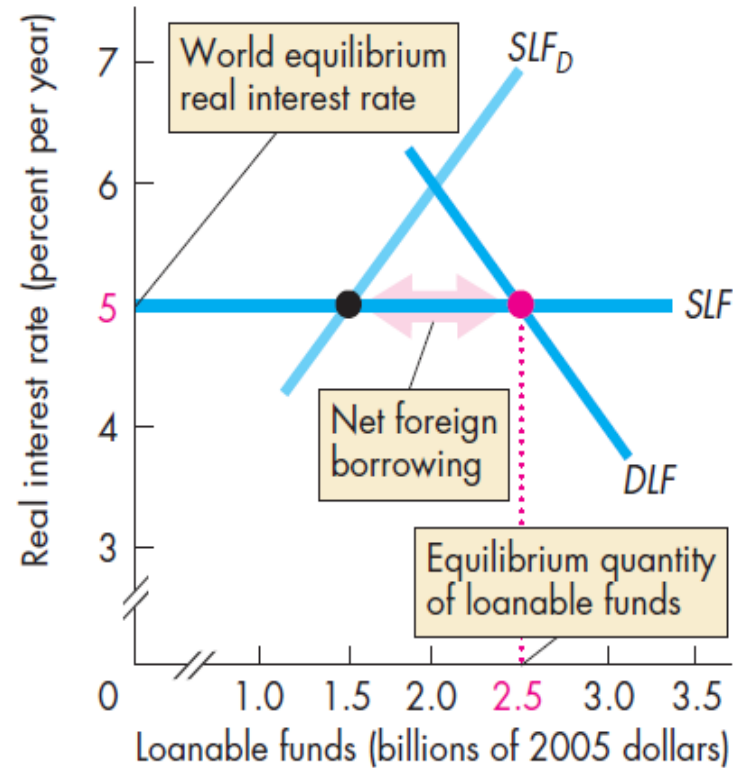
The Global Loanable Funds Market

- A country's loanable funds market connects with the global market through net exports. If a country's net exports are negative ($X < M$), the rest of the world supplies funds to that country and the quantity of loanable funds in that country is greater than national saving. If a country's net exports are positive ($X > M$), the country is a net supplier of funds to the rest of the world and the quantity of loanable funds in that country is less than national saving

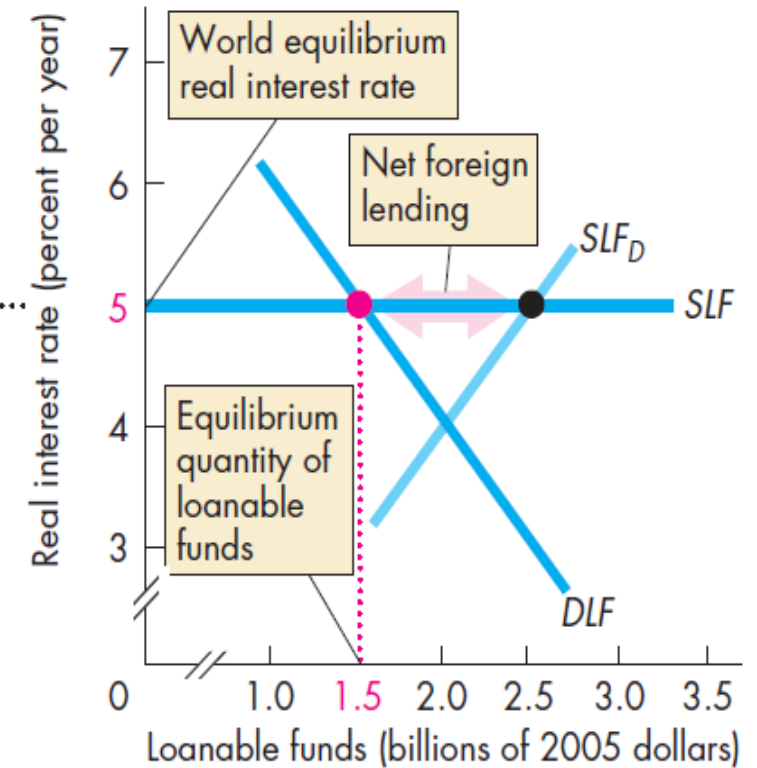
The Global Loanable Funds Market



(a) The global market



(b) An international borrower



(c) An international lender

Question

Q) Suppose that the government of Country X has a budget deficit of \$1 trillion. What do you expect to happen to the real interest rate, the quantity of investment, and the quantity of private saving? Is there any crowding out in this situation?